

Jagraj Gill

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Education

Simon Fraser University – Bachelor of Science in Computing Science

Expected: Dec 2027

Skills

Languages: C, C++, Python, Java, JavaScript, TypeScript, HTML, CSS, x86 Assembly

Frameworks: React, React Native, Node.js, Express.js, REST APIs, GraphQL, FastAPI, CI/CD, Agile

Databases & Tools: Git, Docker, MongoDB, MySQL, SQLite, AWS S3, Linux, Confluence, TCP/IP

Experience

Software Engineer Intern, Savi Finance – Toronto, ON

May 2025 – Dec 2025

- Architected an AI agent orchestration engine and MongoDB schema to autonomously interpret and dispatch work orders to third-party APIs, eliminating manual overhead
- Unified React and React Native feature parity via a mobile-optimized GraphQL schema extension, delivering consistent UX to 1000+ users
- Cut image load failures by 10%, boosting upload reliability for 1000+ users via AWS S3 and GraphQL
- Debugged a login flow locking users out on wrong email entry, adding React handlers for in-place correction

Projects

HTTP Web Server – C++, TCP/IP, HTTP

- Engineered a low-level C++ HTTP server via POSIX sockets, handling TCP connections and serving static HTML, CSS, JavaScript, and image files
- Delivered full GET, POST, and DELETE support with spec-compliant status codes and response headers
- Implemented MIME detection, POST parsing, and error handling for reliable, debuggable request processing
- Enabled concurrent browser and curl client support via poll()-based non-blocking I/O without threads

AI-Powered Study Assistant – Python, JavaScript, FastAPI, Gemini API

- Engineered a PDF-to-flashcard pipeline via PyPDF and Gemini API, automating study material generation and cutting manual prep from hours to seconds
- Enforced a strict JSON schema requiring all flashcard fields, reducing incomplete and malformed AI output
- Built Python validation to reject empty or malformed flashcard content before it reached the UI
- Developed an interactive flashcard UI in JavaScript with flip animations and keyboard navigation

Smart Environmental Monitoring System – C, JavaScript, SQLite, FastAPI

- Built a full-stack IoT monitoring system that streamed live temperature and humidity readings from an embedded sensor device to a web dashboard
- Developed a FastAPI backend that ingested JSON sensor data over Wi-Fi, storing readings in SQLite and powering a live JavaScript dashboard with 24-hour trend views
- Designed fault-tolerant data ingestion that handled connection failures without crashing or failing silently

Portfolio Performance Tracker – Java, Swing

- Built a Java Swing application where users could log stock holdings to visualize portfolio growth trends
- Calculated gain/loss for each stock holding, rendering results as graphs to compare performance across assets

Wireless Robotic Arm – C, ESP32, Embedded Systems

- Built a real-time 4-DOF robotic arm control system using dual ESP32 devices over a peer-to-peer wireless link
- Designed a data pipeline that mapped raw analog sensor readings to per-joint servo angles, applying exponential smoothing and range constraints to prevent erratic or unsafe motion